PROPER PUSHFORWARD PRESERVES RATIONAL EQUIVALENCE OF CYCLES ON GOOD ALGEBRAIC SPACES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-chow.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: a reduction to a single integral closed subspace followed by a three-case analysis, the intermediate case requiring a base change to the generic point, a limit argument and the scheme case

Editorial note. The label lemma-proper-pushforward-rational-equivalence names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Rational equivalence on cycles on a good algebraic space: cycles of dimension k are equivalent when their difference is a locally finite sum of divisors of rational functions on integral closed subspaces of dimension k plus one
- Proper pushforward of cycles, its compatibility with composition of proper morphisms, and the scheme theoretic image of a proper morphism applied to an integral closed subspace
- The scheme case of the statement and the alteration lemma for proper pushforward when the image has the same dimension, together with the criterion identifying a one-dimensional fibre over a generic point with a scheme

Source reference. Original lemma 02S2: sources/stacks-chow/tags.txt, source lines 3645–3645. Prerequisite pointer invoked at proof line(s) 2261; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0ED1: sources/stacks-chow/tags.txt, source lines 17976–17976. Prerequisite pointer invoked at proof line(s) 2253; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0AC8: sources/stacks-chow/tags.txt, source lines 13046–13046. Prerequisite pointer invoked at proof line(s) 2248; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 02R5: sources/stacks-spaces-chow/pinned-original-source.tex, source lines 992–992. Prerequisite pointer invoked at proof

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line(s) 2223; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0EEG: sources/stacks-spaces-chow/pinned-original-source.tex, source lines 820–820. Prerequisite pointer invoked at proof line(s) 2204; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 02RT: sources/stacks-spaces-chow/pinned-original-source.tex, source lines 1819–1819. Prerequisite pointer invoked at proof line(s) 2266; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 02R2: sources/stacks-spaces-chow/pinned-original-source.tex, source lines 908–908. Prerequisite pointer invoked at proof line(s) 2206; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 02QL: sources/stacks-spaces-chow/pinned-original-source.tex, source lines 264–264. Prerequisite pointer invoked at proof line(s) 2177; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 07SR: sources/stacks-chow/tags.txt, source lines 9911–9911. Prerequisite pointer invoked at proof line(s) 2259; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0ADD: sources/stacks-chow/tags.txt, source lines 13086–13086. Prerequisite pointer invoked at proof line(s) 2255; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *In Situation 02QL let $X, Y/B$ be good. Let $p : X \to Y$ be a proper morphism. Suppose $\alpha, \beta \in Z_k(X)$ are rationally equivalent. Then $p_*\alpha$ is rationally equivalent to $p_*\beta$.*

Proof. What do we have to show? Well, suppose we are given a collection

$$i_j : W_j \longrightarrow X$$

of closed immersions, with each W_j integral of δ-dimension $k + 1$ and rational functions $f_j \in R(W_j)^*$. Moreover, assume that the collection $\{i_j(W_j)\}_{j \in J}$ is locally finite on X. Then we have to show that

$$p_* \left(\sum i_{j,*} \mathrm{div}(f_j) \right)$$

is rationally equivalent to zero on X.

Note that the sum is equal to

$$\sum p_* i_{j,*} \mathrm{div}(f_j).$$

Let $W'_j \subset Y$ be the integral closed subspace which is the image of $p \circ i_j$, see Lemma 0EEG. The collection $\{W'_j\}$ is locally finite in Y by Lemma 02R2. Hence it suffices

to show, for a given j, that either $p_*i_{j,*}\mathrm{div}(f_j) = 0$ or that it is equal to $i'_{j,*}\mathrm{div}(g_j)$ for some $g_j \in R(W'_j)^*$.

The arguments above therefore reduce us to the case of a since integral closed subspace $W \subset X$ of δ-dimension $k+1$. Let $f \in R(W)^*$. Let $W' = p(W)$ as above. We get a commutative diagram of morphisms

$$\begin{array}{ccc} W & \xrightarrow{i} & X \\ {\scriptstyle p'}\downarrow & & \downarrow{\scriptstyle p} \\ W' & \xrightarrow{i'} & Y \end{array}$$

Note that $p_*i_*\mathrm{div}(f) = i'_*(p')_*\mathrm{div}(f)$ by Lemma 02R5. As explained above we have to show that $(p')_*\mathrm{div}(f)$ is the divisor of a rational function on W' or zero. There are three cases to distinguish.

The case $\dim_\delta(W') < k$. In this case automatically $(p')_*\mathrm{div}(f) = 0$ and there is nothing to prove.

The case $\dim_\delta(W') = k$. Let us show that $(p')_*\mathrm{div}(f) = 0$ in this case. Since $(p')_*\mathrm{div}(f)$ is a k-cycle, we see that $(p')_*\mathrm{div}(f) = n[W']$ for some $n \in \mathbf{Z}$. In order to prove that $n = 0$ we may replace W' by a nonempty open subspace. In particular, we may and do assume that W' is a scheme. Let $\eta \in W'$ be the generic point. Let $K = \kappa(\eta) = R(W')$ be the function field. Consider the base change diagram

$$\begin{array}{ccc} W_\eta & \longrightarrow & W \\ {\scriptstyle c}\downarrow & & \downarrow{\scriptstyle p'} \\ \mathrm{Spec}(K) & \xrightarrow{\eta} & W' \end{array}$$

Observe that c is proper. Also $|W_\eta|$ has dimension 1: use Decent Spaces, Lemma 0AC8 to identify $|W_\eta|$ as the subspace of $|W|$ of points mapping to η and note that since $\dim_\delta(W) = k+1$ and $\delta(\eta) = k$ points of W_η must have δ-value either k or $k+1$. Thus the local rings have dimension ≤ 1 by Decent Spaces, Lemma 0ED1. By Spaces over Fields, Lemma 0ADD we find that W_η is a scheme. Since $\mathrm{Spec}(K)$ is the limit of the nonempty affine open subschemes of W' we conclude that we may assume that W is a scheme by Limits of Spaces, Lemma 07SR. Then finally by the case of schemes (Chow Homology, Lemma 02S2) we find that $n = 0$.

The case $\dim_\delta(W') = k+1$. In this case Lemma 02RT applies, and we see that indeed $p'_*\mathrm{div}(f) = \mathrm{div}(g)$ for some $g \in R(W')^*$ as desired. □